

■ 3.4.2 Equations in One Variable

The main class of equations that `Solve` and related *Mathematica* functions deal with are *polynomial equations*.

It is easy to solve a linear equation in x .

```
In[1]:= Solve[ a x + b == c , x ]
```

```
Out[1]= {{x -> -(b - c)/a}}
```

One can also solve quadratic equations just by applying a simple formula.

```
In[2]:= Solve[ x^2 + a x + 2 == 0 , x ]
```

```
Out[2]= {{x -> (-a + Sqrt[-8 + a^2])/2},
          {x -> (-a - Sqrt[-8 + a^2])/2}}
```

Mathematica can also find the exact solution to an arbitrary cubic equation.

```
In[3]:= Solve[ x^3 + 34 x + 1 == 0 , x ] [[1]]
```

```
Out[3]= {x ->
```

The results are however often very complicated. Here is the first solution to a comparatively simple cubic equation.

$$\frac{-34}{3 \left(-\frac{1}{2} + \frac{\sqrt{157243}}{6 \sqrt{3}} \right)^{\frac{1}{3}}} + \left(-\frac{1}{2} + \frac{\sqrt{157243}}{6 \sqrt{3}} \right)^{\frac{1}{3}}$$

Mathematica can always find exact solutions to polynomial equations of degree four or less. For cubic and quartic equations, however, the results can be extremely complicated. If the parameters in equations like these are symbolic, there can also be some subtlety in what the solutions mean. The result you get by substituting specific values for the symbolic parameters into the final solution may not be the same as what you would get by doing the substitutions in the original equation.

This generates the first solution to a cubic equation with symbolic parameters.

```
In[4]:= Solve[ x^3 + a x^2 + b x + 2 == 0 , x ] [[1]] ;
```

If you try substituting specific numerical values for the symbolic parameters, you end up dividing by zero, and getting a meaningless result.

```
In[5]:= % /. {a->3, b->3}
Power::infy: Infinite expression 0-(1/3) encountered.
Infinity::indt: Indeterminate expression -(0 ComplexInfinity) encountered.
Out[5]= {x -> Indeterminate}
```

If you use specific values for the parameters in the original equation, then you get the correct result.

```
In[6]:= Solve[ x^3 + 3 x^2 + 3 x + 2 == 0 , x ]
Out[6]= {{x -> -2}, {x ->  $\frac{-1 + \text{Sqrt}[-3]}{2}$ },
          {x ->  $\frac{-1 - \text{Sqrt}[-3]}{2}$ }}
```

In trying to solve polynomial equations with degrees higher than four, *Mathematica* runs into some fundamental mathematical difficulties.

The main mathematical result is that the solutions to an arbitrary polynomial equation of degree five or more cannot necessarily be written as algebraic expressions. More specifically, the solutions cannot be written as combinations of arithmetic functions and k^{th} roots. (It turns out that for equations with degree exactly five, the solutions can in principle be written in a complicated way in terms of elliptic functions; for higher degree equations, even this is not possible.)

There are nevertheless two large classes of higher degree equations which can be solved. The first are those in which the polynomial can be written using **Factor** as product of polynomials with low degrees. The second class are those where the polynomial can be written using **Decompose** as a composition of polynomials with low degrees. When you give a high degree polynomial to **Solve** it successively tries to use **Factor** and **Decompose**, together with some other tricks, to simplify the polynomial. As a result, **Solve** is able to give explicit algebraic solutions to many high degree polynomial equations.

Here is a factorizable polynomial of degree 5.

```
In[7]:= Expand[ Product[ x - i, {i, 5} ] ]
Out[7]= -120 + 274 x - 225 x2 + 85 x3 - 15 x4 + x5
```

Mathematica solves the equation by factoring the polynomial.

```
In[8]:= Solve[ % == 0, x ]
Out[8]= {{x -> 5}, {x -> 4}, {x -> 3}, {x -> 2}, {x -> 1}}
```

`Solve` gives the solution to an equation like this in terms of complex exponentials.

```
In[9]:= Solve[x^6 == 1, x]
Out[9]= {{x -> 1}, {x -> E $\frac{I}{3} \text{Pi}$ }, {x -> E $\frac{2 I}{3} \text{Pi}$ },
{x -> -1}, {x -> E $\frac{4 I}{3} \text{Pi}$ }, {x -> E $\frac{5 I}{3} \text{Pi}$ }}
```

Here is an equation for which explicit algebraic solutions cannot be found. *Mathematica* leaves the result in a symbolic form.

```
In[10]:= Solve[x^5 - 1331 x + 11 == 0, x]
Out[10]= {ToRules[Roots[-1331 x + x^5 == -11, x]]}
```

Applying `N` gives the numerical result.

```
In[11]:= N[%]
Out[11]= {{x -> -6.04217},
{x -> -0.00206611 - 6.04011 I},
{x -> -0.00206611 + 6.04011 I}, {x -> 0.00826446},
{x -> 6.03804}}
```

Here is a complicated polynomial.

```
In[12]:= p = 3 + 3 x - 7 x^2 - x^3 + 2 x^4 +
3 x^7 - 3 x^8 - x^9 + x^10
Out[12]= 3 + 3 x - 7 x^2 - x^3 + 2 x^4 + 3 x^7 - 3 x^8 - x^9 +
x^10
```

Mathematica finds some of the solutions, and leaves the rest in a symbolic form. Applying `N` would give the numerical solutions.

```
In[13]:= Solve[% == 0, x]
Out[13]= {{x -> 1}, {x -> Sqrt[3]}, {x -> -Sqrt[3]},
ToRules[Roots[1 + 2 x + x^7 == 0, x]]}
```

If you ultimately need only the numerical solution, it is usually a lot faster to use `NRoots` to ask for numerical results from the outset.

```
In[14]:= NRoots[p == 0, x]
Out[14]= x == -1.73205 || x == -0.868688 - 0.585282 I ||
x == -0.868688 + 0.585282 I || x == -0.496292 ||
x == 0.0763556 - 1.14095 I ||
x == 0.0763556 + 1.14095 I || x == 1. ||
x == 1.04048 - 0.56735 I ||
x == 1.04048 + 0.56735 I || x == 1.73205
```

If there are approximate numbers in your original equation, `Solve` will immediately get a numerical solution.

```
In[15]:= Solve[x^3 + 7.8 x + 1 == 0, x]
Out[15]= {{x -> -0.127937}, {x -> 0.0639683 - 2.79504 I},
{x -> 0.0639683 + 2.79504 I}}
```

If **Solve** cannot find an algebraic solution to a high degree polynomial equation, then it is a good guess that no such solution exists. However, you should realize that one can construct complicated equations that have algebraic solutions which the procedures built into *Mathematica* do not find. The simplest example of such an equation that we know is $-23 - 36x + 27x^2 - 4x^3 - 9x^4 + x^6 = 0$, which has a solution $x = 2^{\frac{1}{3}} + 3^{\frac{1}{2}}$ that **Solve** does not find.

When *Mathematica* can find solutions to an n^{th} degree polynomial equation, it always gives exactly n solutions. The number of times that each root of the polynomial appears is equal to its multiplicity.

Solve gives two identical solutions to this equation.

```
In[16]:= Solve[(x-1)^2 == 0, x]
Out[16]= {{x -> 1}, {x -> 1}}
```

Mathematica knows how to solve some equations which are not explicitly in the form of polynomials.

Here is an equation that is not explicitly of polynomial form.

```
In[17]:= Solve[ Sqrt[1-x] + Sqrt[1+x] == a, x ]
Out[17]= {{x ->  $\frac{a \sqrt{4 - a^2}}{2}$ }, {x ->  $\frac{-(a \sqrt{4 - a^2})}{2}$ }}
```

Mathematica can always give you numerical approximations to the solutions of a polynomial equation. For more general equations, involving say transcendental functions, there is often no systematic procedure even for finding numerical solutions. Section 3.9.6 discusses approaches to this problem in *Mathematica*.

This finds a numerical solution to the equation $x \sin(x) = \frac{1}{2}$, close to $x = 1$.

```
In[18]:= FindRoot[ x Sin[x] - 1/2 == 0 , {x, 1} ]
Out[18]= {x -> 0.740841}
```

Plotting a graph of $x \sin(x) - \frac{1}{2}$ makes it pretty clear that there are in fact an infinite number of solutions to the equation.

